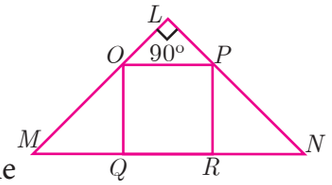
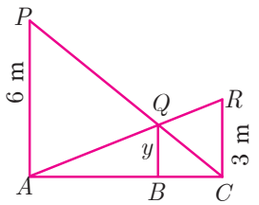
7. If figure $OPRQ$ is a square and $\angle MLN = 90°$. Prove that (i) $\Delta LOP \sim \Delta QMO$ (ii) $\Delta LOP \sim \Delta RPN$ (iii) $\Delta QMO \sim \Delta RPN$ (iv) $QR^2 = MQ \times RN$



8. If $\Delta ABC \sim \Delta DEF$ such that area of ΔABC is 9cm^2 and the area of ΔDEF is 16cm^2 and $BC = 2.1$ cm. Find the length of EF.

9. Two vertical poles of heights 6 m and 3 m are erected above a horizontal ground AC. Find the value of y.



10. Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{2}{3}$ of the corresponding sides of the triangle PQR (scale factor $\frac{2}{3} < 1$).

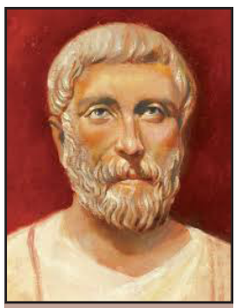
11. Construct a triangle similar to a given triangle LMN with its sides equal to $\frac{4}{5}$ of the corresponding sides of the triangle LMN (scale factor $\frac{4}{5} < 1$).

12. Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC (scale factor $\frac{6}{5} > 1$).

13. Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle PQR (scale factor $\frac{7}{3} > 1$).

4.3 Thales Theorem and Angle Bisector Theorem

4.3.1 Introduction



Thales
(640 - 540 BC (BCE))

Thales, (640 - 540 BC (BCE)) the most famous Greek mathematician and philosopher lived around seventh century BC (BCE). He possessed knowledge to the extent that he became the first of seven sages of Greece. Thales was the first man to announce that any idea that emerged should be tested scientifically and only then it can be accepted. In this aspect, he did great investigations in mathematics and astronomy and discovered many concepts. He was credited for providing first proof in mathematics, which today is called by the name "Basic Proportionality Theorem". It is also called "Thales Theorem" named after its discoverer.

The discovery of the Thales theorem itself is a very interesting story. When Thales travelled to Egypt, he was challenged by Egyptians to determine the height of one of several magnificent pyramids that they had constructed. Thales accepted the challenge and used similarity of triangles to determine the same successfully, another triumphant application of Geometry. Since X_0, X_1 and H_0 are known, we can determine the height H_1 of the pyramid.

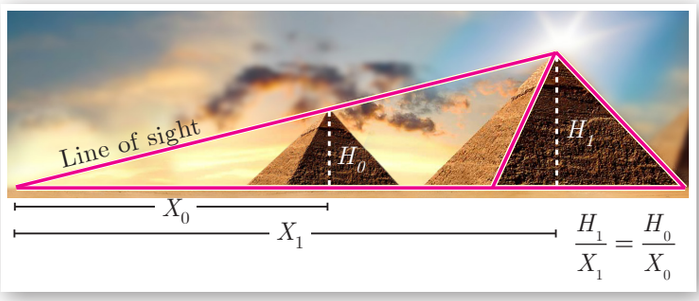


Fig. 4.27

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